

Youngseok Lee

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Education

Sungkyunkwan University (SKKU), Nano Engineering

- GPA 4.22 / 4.5 (major GPA 4.31 / 4.5); Dean's List, Fall 2022 and Spring 2023.
- Relevant courses: Quantum Mechanics I, Quantum Physics and Quantum Information, Quantum Engineering, Understanding Quantum Computers, Introduction to Solid State Physics, Computational Simulation of Material Properties, Engineering Numerical Analysis, AI and Digital Mathematics.

Suwon, Republic of Korea
Mar 2018 – Aug 2024

Experience

NORMA Inc., Quantum AI Developer, Quantum AI Team

Research and development of quantum machine learning models and the classical simulation tools used to train them.

- Co-developed and maintain PADO-Pauli, a GPU-accelerated differentiable Pauli-propagation engine for large-scale circuit simulation and training (40–120 qubits), shipped as the Python package padopauli.
- Proposed and analyzed mixed IQP-QCBMs, ancilla-extended IQP generative models with provable barren-plateau avoidance and a cluster initialization scheme; deployed 42-qubit mixed IQP Born machines on the Quantinuum H2 trapped-ion quantum computer.
- Developed SAFE ma-QAOA, a surrogate-assisted, parameter-distilled multi-angle QAOA pipeline with exact fine-tuning (QCE26 oral).
- Applied QGAN and IQP-QCBM models to molecular generation for KRAS G12D inhibitor candidates, with QAOA experiments on IonQ Forte.
- Contributed to an in-house quantum SDK (circuit representation, user API, circuit output and visualization).

Seoul, Republic of Korea

Qunova Computing, Research Intern

- Designed a tensor-network-based machine learning model for drug toxicity prediction.

Daejeon, Republic of Korea
Oct 2023 – Feb 2024
5 months

Superconducting Quantum Computing Lab (Prof. Yonuk Chong), Sungkyunkwan University, Undergraduate Researcher

- Pulse-level control of superconducting qubits, including gate implementation and calibration experiments.
- Superconducting qubit measurement and characterization, such as qubit frequency and coherence-time measurements.

Suwon, Republic of Korea
Mar 2021 – Oct 2023
2 years 8 months

Publications

Trainability and Mode Separation of Mixed IQP-QCBMs

Youngseok Lee, Hyunwoo Kim
arxiv.org/abs/2607.27883

SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation

Youngseok Lee, Hyunwoo Kim

PADO-Pauli: A GPU-Accelerated Implementation of Pauli Propagation

Hyunwoo Kim, Youngseok Lee
github.com/Norma-Q/PADO-Pauli

Conference Presentations

- **SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation.** Oral presentation, QALG track, IEEE Quantum Week 2026 (QCE26), Toronto, Canada, September 13–18, 2026 (scheduled).
- **PADO-Pauli.** Poster, IEEE Quantum Week 2026 (QCE26), Toronto, Canada, September 13–18, 2026 (scheduled).
- **Trainability and Mode Separation of Mixed IQP Circuits.** Contributed talk, [ML4QT Symposium](#), Institute for Quantum Computing, University of Waterloo, September 23–25, 2026 (scheduled).
- **PADO-Pauli: A GPU-Accelerated Implementation of Pauli Propagation.** Poster, Conference on Quantum Information (CQI 2026), Seoul, Republic of Korea, July 2026.
- **Two Approaches of Potential Quantum Advantage with Quantum Generative Models in Drug Discovery.** Poster, QISK Annual Meeting 2026, Seoul, Republic of Korea, February 2026.

Patents

- **Quantum Computing Method Using a Pauli Graph Generated by Zero Filtering to Perform Surrogate Operations and Quantum Computing System Performing the Same.** Korean patent, application no. 10-2026-0070495 (filed 2026-04-20), registered. Applicant: NORMA Inc.
- **Quantum Artificial Intelligence Training Method Performing Surrogate Training Using Pauli Graphs and Quantum Artificial Intelligence Training System Performing the Same.** Korean patent, application no. 10-2026-0070494 (filed 2026-04-20), registered. Applicant: NORMA Inc.

Honors and Awards

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| Finalist, Pasqal Quantum AI for Sustainability Challenge
Selected as one of 15 finalist teams out of 137 global submissions for a renewable-energy portfolio optimization framework using quantum-classical reinforcement learning.
Team Qorea | Mar 2025 |
| Center Director's Award, 2024 Quantum Information Competition (Hackathon)
Applied Neural Quantum Embedding (NQE) to time-series learning by combining it with a quantum GRU (QGRU), outperforming a classical GRU baseline in weather-forecasting accuracy.
Team Qurious | June 2024 |
| Honorable Mention, ACM/IEEE Quantum Computing on Drug Discovery Challenge, ICCAD 2023
Top 8 of 73 registered teams; ranked 1st in accuracy (99.96, highest among 22 submitting teams) and 4th overall, using qubit tapering and reference-state error mitigation (REM) for ground-state energy estimation.
Team Quantum Korea | Oct 2023 |
| SKKU President's Award, 2023 Quantum Information Competition (Hackathon)
Applied reference-state error mitigation to Hamiltonian simulation on quantum hardware, exploiting the time-invariance of a known ground state under Hamiltonian evolution to estimate and correct hardware errors for states evolved near the reference.
Team Minchodan | June 2023 |

Research Mentoring

- Supervised research intern Dohyun Kim (January 19 – April 19, 2026) on *Reinforcement Learning for Adaptive Pauli Propagation*, and wrote a letter of recommendation.

Skills

Quantum computing

Programming and HPC

Languages _____

Korean
Native

English
Fluent

Interests _____

Research interests